# Utilizing Semantic Link Labs to proactively identify issues in models and reports

Jason Romans

Power BI & Fabric SUMMIT 2026

Special thanks to Fabric and Power BI Team at

Microsoft

This Summit presented to you by

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# Jason Romans

Cloud Data & Integration Developer

## The DAX Shepherd

Microsoft® MVP Most Valuable Professional

@sql_jar

jason-r-sql-jar

https://thedaxshepherd.com

📍 Nashville, TN, USA

🛠️ Began Career as a SQL Server DBA

➡️ Transitioned to Microsoft BI Stack

📦 Data Engineering to Data Modeling

✒️ Infrequent Blogger

🧭 Fan of Dimensional Models & Doctor Who

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# Agenda

1. Intro
2. Notebooks
3. Semantic Link
4. Semantic Link Labs
5. Conclusion

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# Why Semantic Link Labs?

# Administration

Automate semantic model deployment, migration, and governance tasks across Fabric workspaces

# Maintenance

Optimize semantic models with Vertipaq analysis and maintain Fabric lakehouses through automated table management

# Issue Detection

Identify data quality problems, orphaned objects, and performance bottlenecks before they impact users

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# Requirements

- Fabric Capacity
- Fabric Notebook

# Agenda

1. Intro
2. **Notebooks**
3. Semantic Link
4. Semantic Link Labs
5. Conclusion

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# Compute

- Pure Python Notebook
  - 1 Node VM (Scale Up)
- Spark Notebook
  - Cluster (Scale Out)

# Type of Notebook

## Notebook requirements vary by function:

- “This function can be executed in either a PySpark or pure Python notebook.”
- “This function is only available in a PySpark notebook.”

# Agenda

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3. **Semantic Link**
4. Semantic Link Labs
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# Missing Link

Semantic link is a feature that allows you to establish a connection between semantic models and Synapse Data Science in Microsoft Fabric.

-- Reference: https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview

# All the Things

Semantic link is a feature that allows you to establish a connection between semantic models and Synapse Data Science in Microsoft Fabric.

**semantic models**

Models, Reports, Lakehouse, Workspaces and more

**Synapse Data Science**

Fabric Notebook – Apache Spark-based and pure Python notebook environments

# A Tale of Two Links

Both are Available Only in Microsoft Fabric

**Semantic Link**

- Base
- Driver or API
- Included in the default runtime for the current version

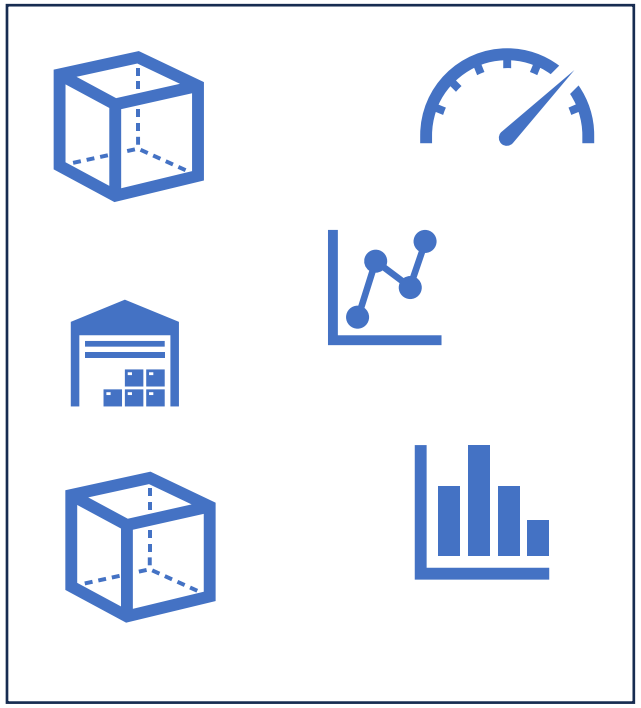
# A Link Between Worlds

**Semantic Link Labs**

- Michael Kovalsky
  - “Kovalsky’s Laboratory”
- “Expansion Pack” -- Kurt Buhler
- Uses Semantic Link
  - import sempy.fabric as fabric
- Open source – GitHub Repository
  - Active Development

# Semantic Link Labs

Microsoft Fabric



Semantic Link

List Tables

List Workspaces

List Models

List Reports

Semantic Link Labs

Migrate to Direct Lake

Best Practice Analyzer

Manage Reports

View Broken Reports

Rebind Reports

# Installing

- Semantic Link
  - Spark 3.4 in default runtime
    - Update to newest version
      - %pip install –U semantic-link
  - Spark 3.3 or below – need to install
    - %pip install –U semantic-link

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# What Version of Spark

**Workspace settings**

- General
- License info
- Azure connections
- System storage
- Git integration
- OneLake
- Workspace identity
- Network security

Power BI

Delegated Settings

- OneLake settings

Data Engineering/Science

- Spark settings

Data Factory

## Spark settings

Configure and manage settings for Spark workloads and the default environment for the workspace.

Pool | **Environment** | High concurrency | Automatic log

**Set default environment** Off

The default environment will provide Spark properties, libraries, and developer settings for notebooks and Spark job definitions in this workspace when users don't select a different environment. Learn more about Set default environment

Runtime

**Runtime Version**

Runtime version defines which version of Spark your Spark pool will use. Learn more about Runtime Version

1.2 (Spark 3.4, Delta 2.4)

# Spark Version Command

## spark.version

```
1  spark.version
```

✓ <1 sec - Command executed in 260 ms by Jason Romans on 2:00:03 PM, 10/08/24

```
'3.4.3.5.3.20240904.5'
```

# What Version of Semantic Link

%pip show semantic-link

or %pip list | grep semantic-link

```
1   %pip show semantic-link
```

[23] ✓ 4 sec - Command executed in 3 sec 588 ms by Jason Romans on 11:49:10 PM, 10/19/24 PySpark (Python)

```
Name: semantic-link
Version: 0.8.1
Summary: Semantic link for Microsoft Fabric
Home-page: https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview
Author: Microsoft
Author-email: semanticdatascience@service.microsoft.com
License: proprietary and confidential
Location: /nfs4/pyenv-dd0ba783-2069-4fdc-8c80-bc03c74db705/lib/python3.11/site-packages
Requires: semantic-link-functions-geopandas, semantic-link-functions-holidays, semantic-link-functions-meteostat, semantic-link-
functions-phonenumbers, semantic-link-functions-validators, semantic-link-sempy
Required-by:
Note: you may need to restart the kernel to use updated packages.
```

# Import the Module

```
# Give it a friendly name – think Alias
import sempy.fabric as fabric
```

# Done in a Notebook

Text Descriptions - Markdown

Code - Python

## Install the latest .whl package

Check here to see the latest version.

```
%pip install semantic-link-labs
```

- Session ready in 8 sec 603 ms. Command executed in 26 sec 202 ms by Jason Romans on 10:29:04 AM, 10/23/24

## Install Wheel from File

```
%pip install /lakehouse/default/Files/semantic_link_labs-0.8.3-py3-none-any.whl
```

- Command executed in 22 sec 745 ms by Jason Romans on 10:07:06 AM, 10/23/24

## Show Semantic Link and Labs installed

```
%pip show semantic-link-sempy
print('\n')
%pip show semantic-link-labs
```

✓ - Command executed in 6 sec 382 ms by Jason Romans on 10:30:20 AM, 10/23/24

```
Name: semantic-link-sempy
Version: 0.8.1
Summary: Semantic link for Microsoft Fabric
Home-page: https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview
Author: Microsoft
```

# Uses Pandas Ecosystem (DataFrame)

Knowledge of working with Pandas DataFrame helpful

DataFrame – data table

# List Semantic Models

```
import sempy.fabric as fabric
fabric.list_datasets()
```

Dataset = Semantic Model

| | Dataset Name | Dataset ID | Created Timestamp | Last Update |
|---|---|---|---|---|
| 0 | Contoso10K | 47c34560-ef4d-46c6-825e-20cb9f11ba9d | 2023-05-04 14:36:12 | NaT |
| 1 | FabSLL_Lakehouse | e6d18d3a-b407-4fb3-813a-418b76388b11 | 2021-02-12 23:00:58 | NaT |
| 2 | DataflowsStagingLakehouse | 7c76f16d-2364-4c33-89bb-6960ac29cb5d | 2021-02-12 23:00:58 | NaT |
| 3 | DataflowsStagingWarehouse | 625cb1b9-de7e-425f-90b3-727ac87268db | 2021-02-12 23:00:58 | NaT |
| 4 | Contoso10K_DL | 261101fb-fc2d-4511-be34-1def1b4530fe | 2019-09-17 05:50:29 | NaT |
| 5 | Contoso10K_DLL1 | 40b52877-7e62-43cd-9a38-7c09b62f9048 | 2019-09-17 05:50:29 | NaT |
| 6 | Contoso10K_ABC | cc93c0f1-e112-4adc-a009-3fb3a7a12f3b | 2019-09-17 05:50:29 | NaT |
| 7 | Contoso10K_SLL | e45a6ce5-5ea5-4b87-ad88-136dc4cabc27 | 2019-09-17 05:50:29 | NaT |
| 8 | Contoso10K_SL1 | 421040eb-ba74-4304-bb92-708c1667eb61 | 2019-09-17 05:50:29 | NaT |
| 9 | Contoso10K_SL2 | b84b790d-04a2-406d-ba17-2191186ebafa | 2019-09-17 05:50:29 | NaT |
| 10 | Contoso10K_SL9 | 5ca86b73-bef1-43d6-af09-5c12e358d391 | 2019-09-17 05:50:29 | NaT |

# List Tables

```
tables = fabric.list_tables(workspace="ws", dataset = "model")
display(tables)
```

| | Name | Description | Hidden | Data Category | Type |
|---|---|---|---|---|---|
| 0 | Customer | | False | | Table |
| 1 | Sales | | False | | Table |
| 2 | Date | | False | | Table |
| 3 | Store | | False | | Table |
| 4 | Product | | False | | Table |

# List Workspaces

```
fabric.list_workspaces()
```

✓ - Command executed in 806 ms by Jason Romans on 5:25:32 PM, 10/08/24 PySpark (Python)

| | Id | Is Read Only | Is On Dedicated Capacity | Capacity Id | Default Dataset Storage Format | Type | Name |
|---|---|---|---|---|---|---|---|
| 0 | bfab8dff-bdfc-4943-9996-7dc97a4e4d38 | False | False | NaN | NaN | Workspace | JAXSQL2023 |
| 1 | 41e69008-f6c3-42c0-8c0a-739f1a7a9a0a | False | True | 3b9ac229-bcbb-4aa4-8543-72b6db25e330 | Small | Workspace | ONSQL2023 |
| 2 | 10c0ad3f-7a0b-4c0e-8b7d-f4a6170c5dde | False | True | 3b9ac229-bcbb-4aa4-8543-72b6db25e330 | Small | Workspace | SOFLSQL2023 |
| 3 | ee7a8d30-f109-4848-9db3-be711a0f24d4 | False | False | NaN | NaN | AdminInsights | Admin monitoring |
| 4 | 10c2bd0a-5d3d-4e77-883c-f16af027fcd5 | False | True | 3b9ac229-bcbb-4aa4-8543-72b6db25e330 | Small | Workspace | COLSQLSAT2023 |
| 5 | 365a3880-83f7-4014-bad3-a006c34e2bb1 | False | True | 3b9ac229-bcbb-4aa4-8543-72b6db25e330 | Small | Workspace | BRSQLSAT2023 |
| 6 | e6e71dc2-2a8e-482a-8bc9-ff5c16ed0311 | False | True | 3b9ac229-bcbb-4aa4-8543-72b6db25e330 | Small | Workspace | DENSQLSAT2023 |

# List Workspaces on Dedicated Capacity

```
ws = fabric.list_workspaces()
dedicated = ws[ws["Is On Dedicated Capacity"] == True]
display(dedicated)
```

✓ - Command executed in 790 ms by Jason Romans on 5:37:07 PM, 10/08/24 PySpark (Python)

Table | Chart | Download | Showing rows 1 - 22 | Inspect | Search

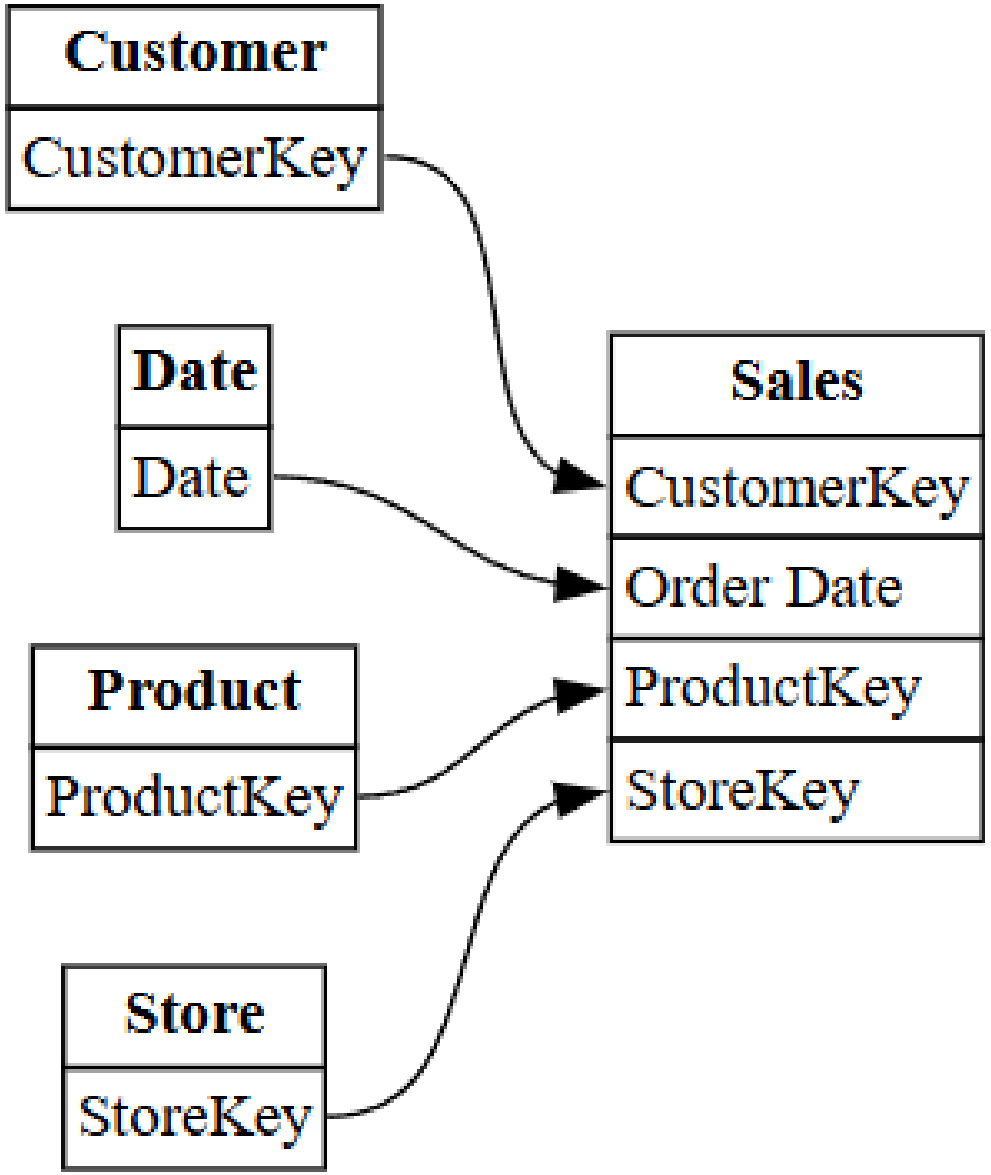
| | Id | Is Read Only | Is On Dedicated Capacity | Capacity Id | Default Dataset Storage Format | Type | Name |
|---|---|---|---|---|---|---|---|
| 1 | 41e69008-f... | false | true | 3b9ac229-bcbb... | Small | Workspace | ONSQL2023 |
| 2 | 10c0ad3f-7... | false | true | 3b9ac229-bcbb... | Small | Workspace | SOFLSQL2023 |
| 3 | 10c2bd0a-... | false | true | 3b9ac229-bcbb... | Small | Workspace | COLSQLSAT2023 |
| 4 | 365a3880-... | false | true | 3b9ac229-bcbb... | Small | Workspace | BRSQLSAT2023 |
| 5 | e6e71dc2-... | false | true | 3b9ac229-bcbb... | Small | Workspace | DENSQLSAT2023 |
| 6 | 782b40a8-... | false | true | 3b9ac229-bcbb... | Small | Workspace | SQLMAB |
| 7 | 62e33fa7-f... | false | true | 3b9ac229-bcbb... | Small | Workspace | SQLSAT_Denver2023 |
| 8 | 9a5d4f1a-b... | false | true | 3b9ac229-bcbb... | Small | Workspace | MNSQLSAT2023 |
| 9 | ba1e683d-... | false | true | 3b9ac229-bcbb... | Small | Workspace | ORLANDOSQLSAT2023 |
| 10 | be75ac57-... | false | true | 3b9ac229-bcbb... | Small | Workspace | SV-SQLSat2023 |
| 11 | 4d139fc8-0... | false | true | 3b9ac229-bcbb... | Small | Workspace | ORWA23_SQLSAT |
| 12 | 23c86de6-... | false | true | 3b9ac229-bcbb... | Large | Workspace | TE Training |
| 13 | ad0afb41-... | false | true | 3b9ac229-bcbb... | Small | Workspace | PremTest |
| 14 | 2e6ae09d-... | false | true | 3b9ac229-bcbb... | Large | Workspace | SQLBits2024 |
| 15 | 5f6564c6-1... | false | true | 27046857-fc30-... | Small | Workspace | MetaDriven-2 |

Selected Cell 14 of 21 ce

# Relationships

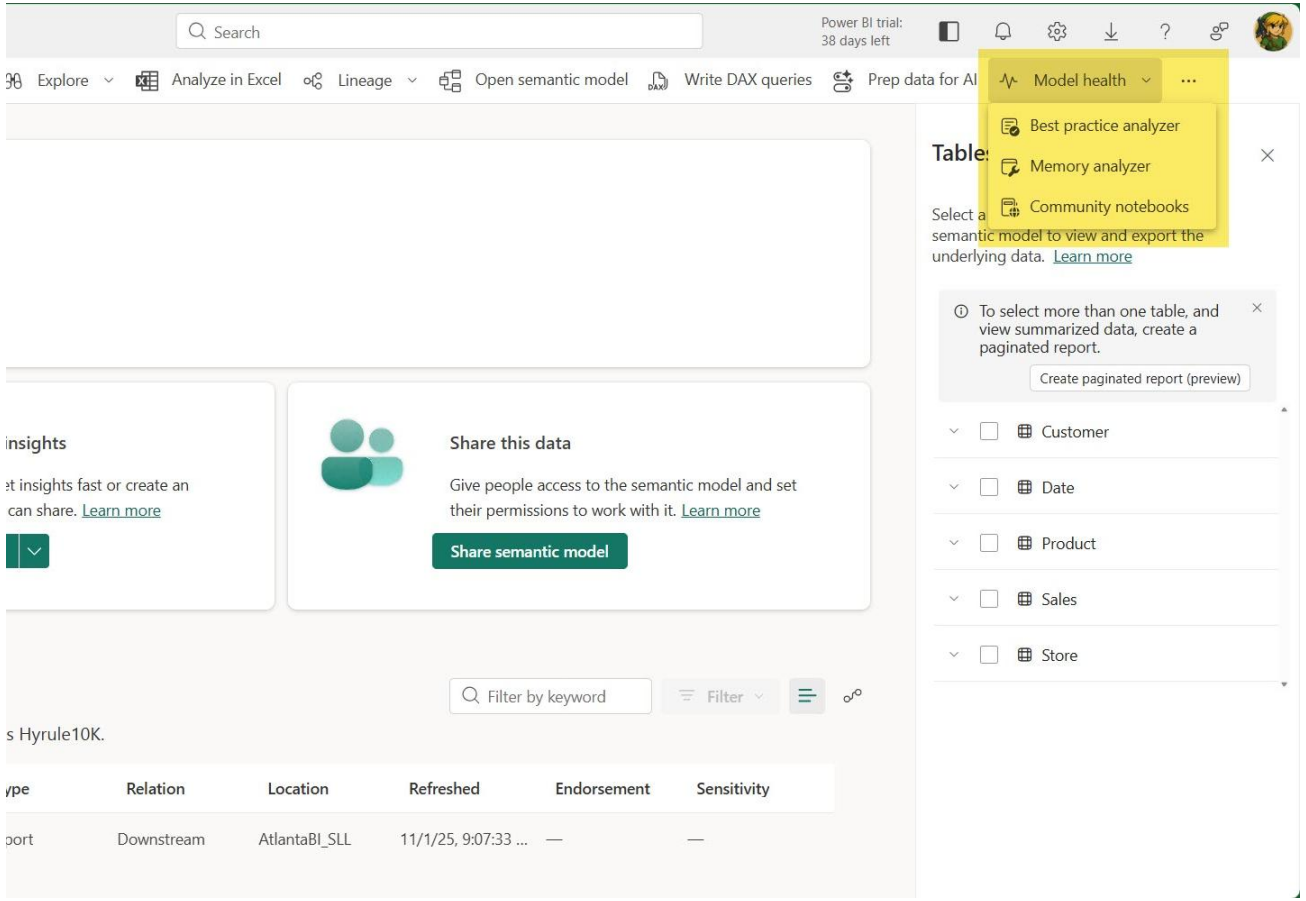
```
from sempy.relationships import
plot_relationship_metadata as prm

prm(
    fabric.list_relationships(
        workspace="Summit2025_SLL",
        dataset="Hyrule10K"
    )
)
```



# Model Health in Microsoft Fabric

1. Best Practice Analyzer
2. Memory Analyzer
3. Community Notebooks



# Semantic Link Demo

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# Semantic Link Labs

- Standard Python library
- Completely free and open-source
- Backed by Microsoft (GitHub repository)
- Continuous development and improvements

# Origin Story

Migrations:
Semantic Models from
Import to Direct Lake

# Active Development

June 18, 2024 at 0.4.2

Now up to ~~0.12.8~~

~~0.12.9~~

0.12.10

# Install the Package

```
# Install Semantic Link Labs
%pip install semantic-link-labs

# Import module with a shorter name
# Easier to Type
import sempy_labs as labs
```

# Notebooks to Get you Started

- Migration to Direct Lake
- Best Practice Analyzer Report
- Capacity Migration
- Delta Analyzer
- Model Optimization
- Query Scale Out

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# Even More Helper Notebooks

- Report Analysis
- SQL
- Semantic Model Management
- Semantic Model Refresh
- Service Principal
- Tabular Object Model

# Obstacles

- May Require Spark
  - This function may be executed in either a PySpark or pure Python notebook. If executing in a pure Python notebook, the DataFrame must be a pandas DataFrame.
- May not fully work with Lakehouse Schemas

# Update Dependencies

- Installing Semantic Link Labs will update Semantic Link

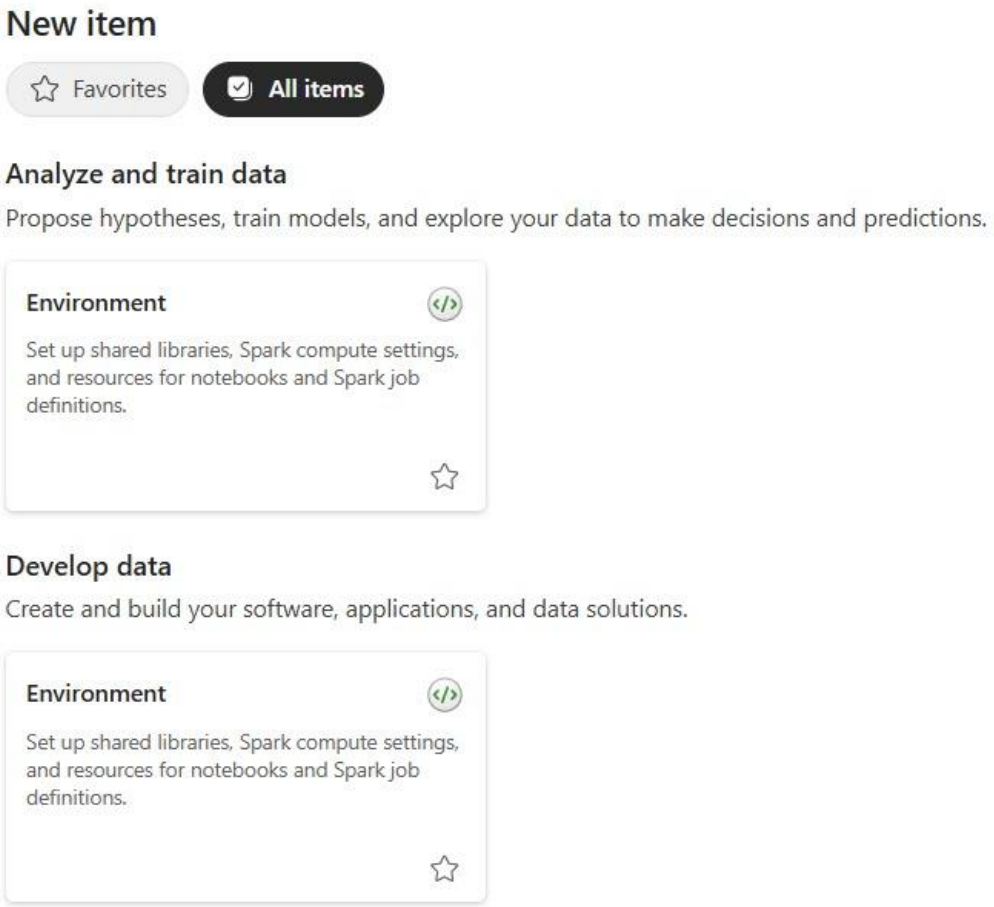
# Expands Semantic Link

- Does not replace or overwrite
- Have both available
- You will use both
- Just like installing DuckDB or other packages

# Environment

- Install shared libraries
- Spark settings
- Not available for Pure Python Notebook

# Create Environment

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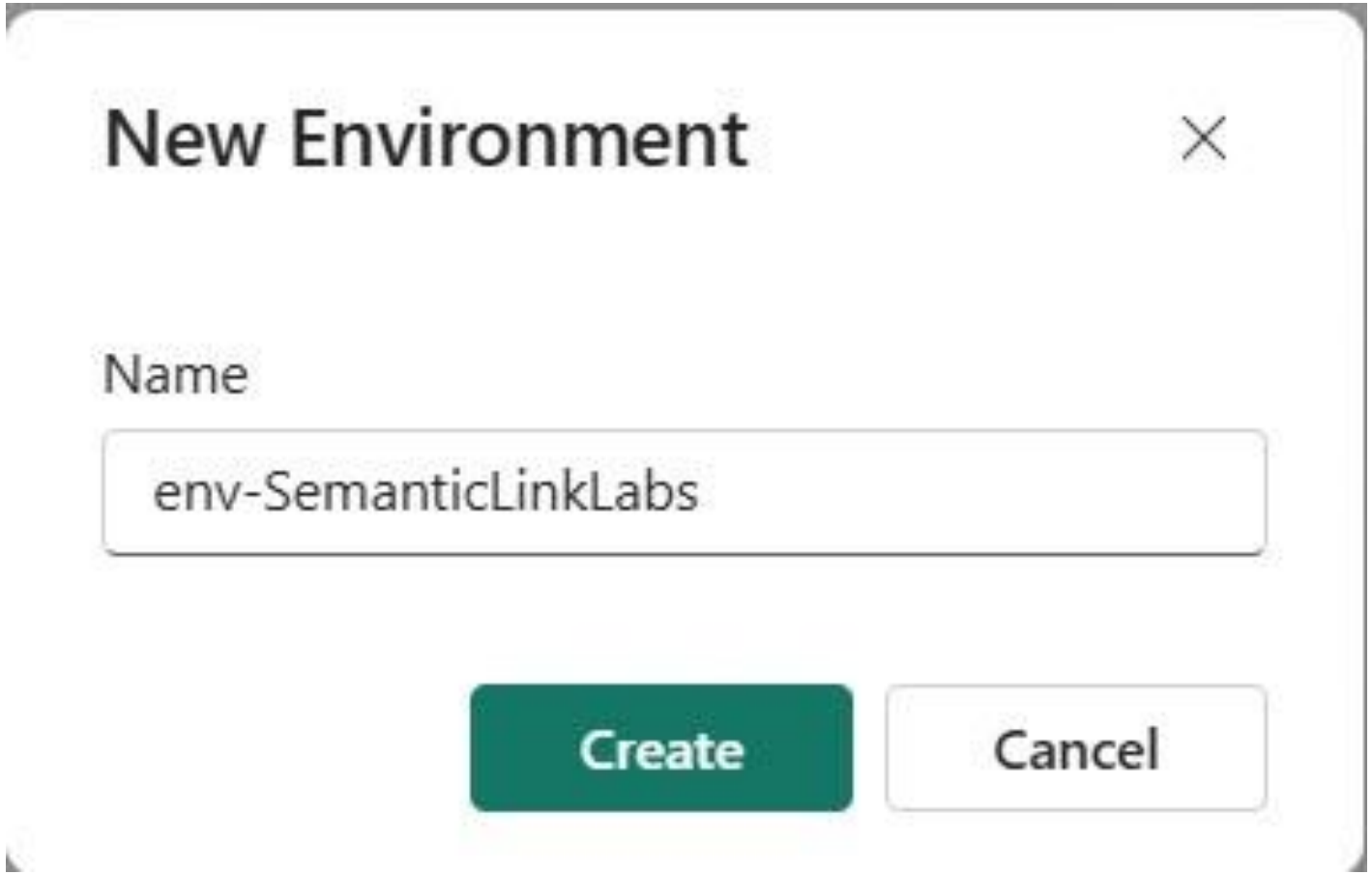
# Name Environment

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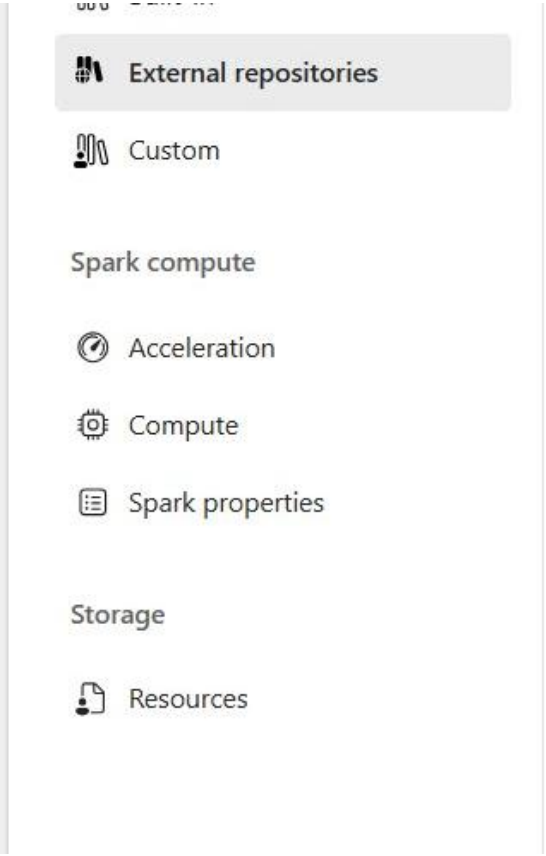
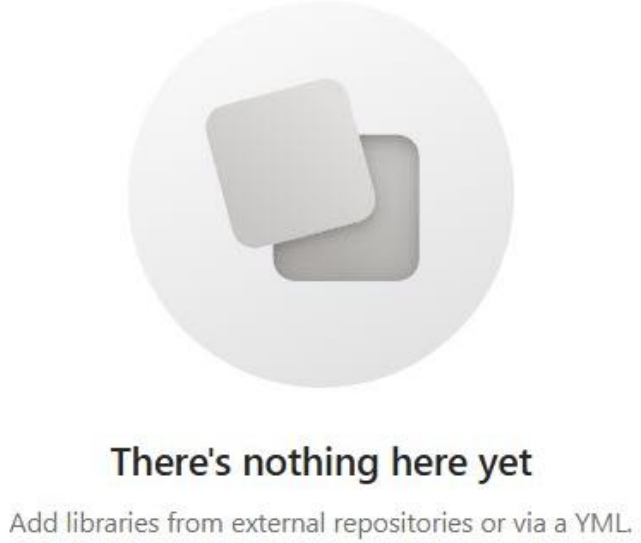
# Add Library

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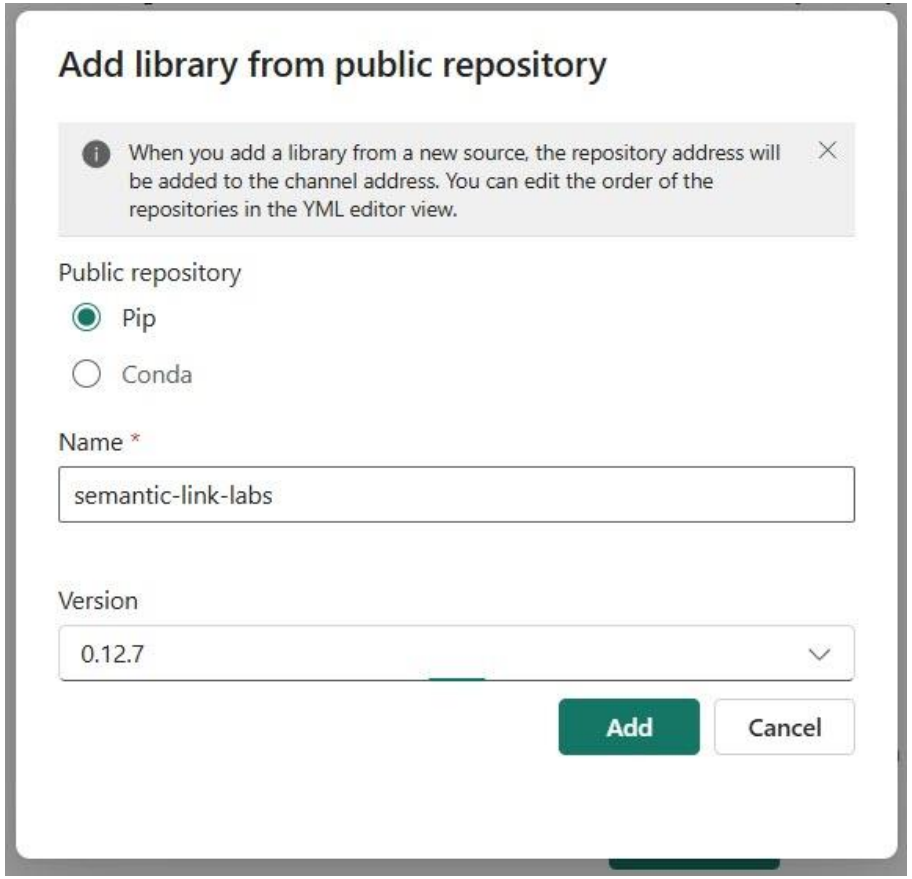
# Add Library from Repo (Choice)

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# Specific Versions

- Choose your development version
- Control your upgrade path

# List of Libraries

You have unpublished changes. To apply these changes to notebooks and Spark job definition run in this environment, select Publish. To save your changes without updating the environment, select Save.

Libraries

- Built-in
- External repositories 1
- Custom

Spark compute

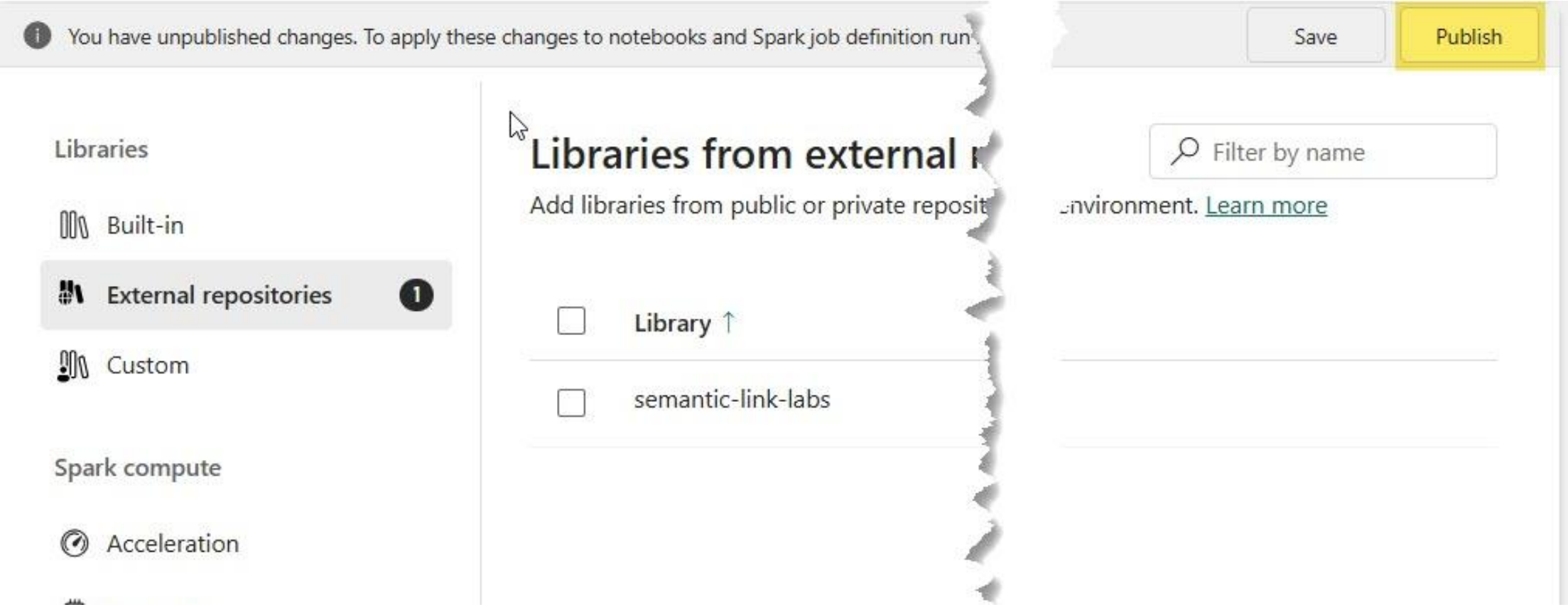
- Acceleration
- Compute
- Spark properties

## Libraries from external repositories

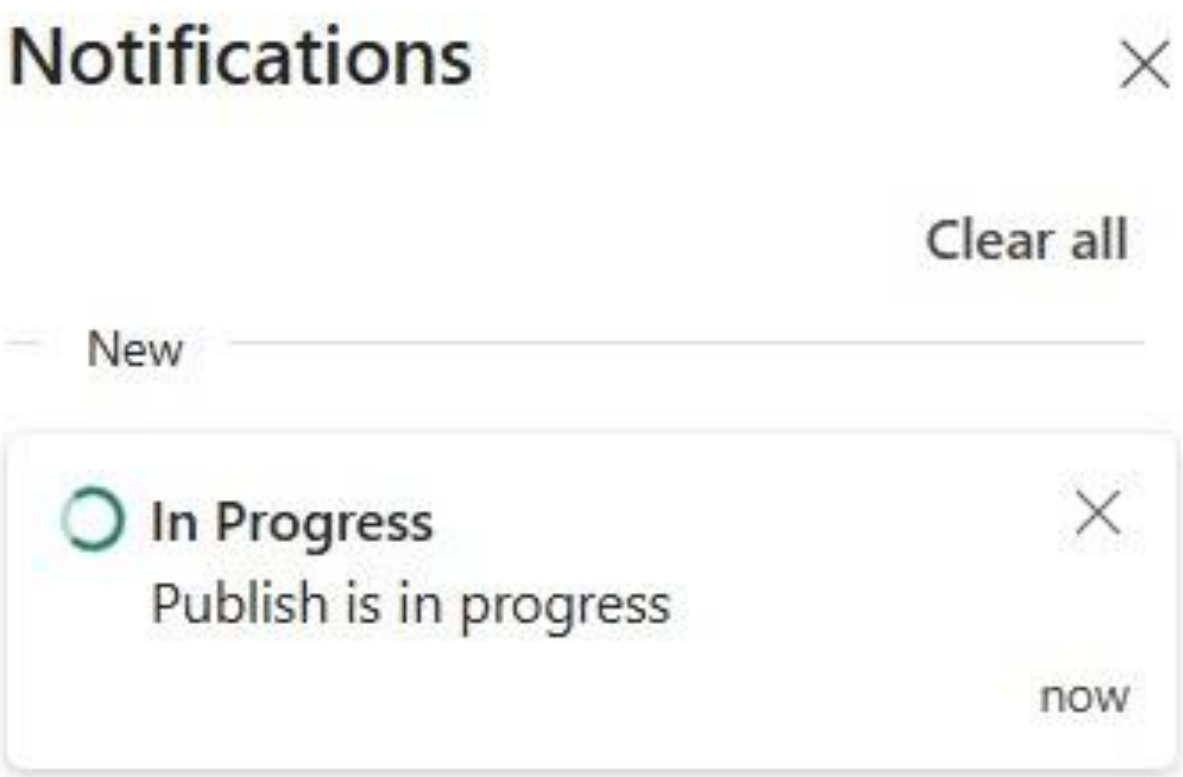
Add libraries from public or private repositories using the list view or via YML. Added libraries will be available if you run your notebook or Spark job

| Library ↑ | Version | Status | Last updated |
|---|---|---|---|
| semantic-link-labs | 0.12.7 | New | New |

# Publish (Important)



# Environment takes time to publish

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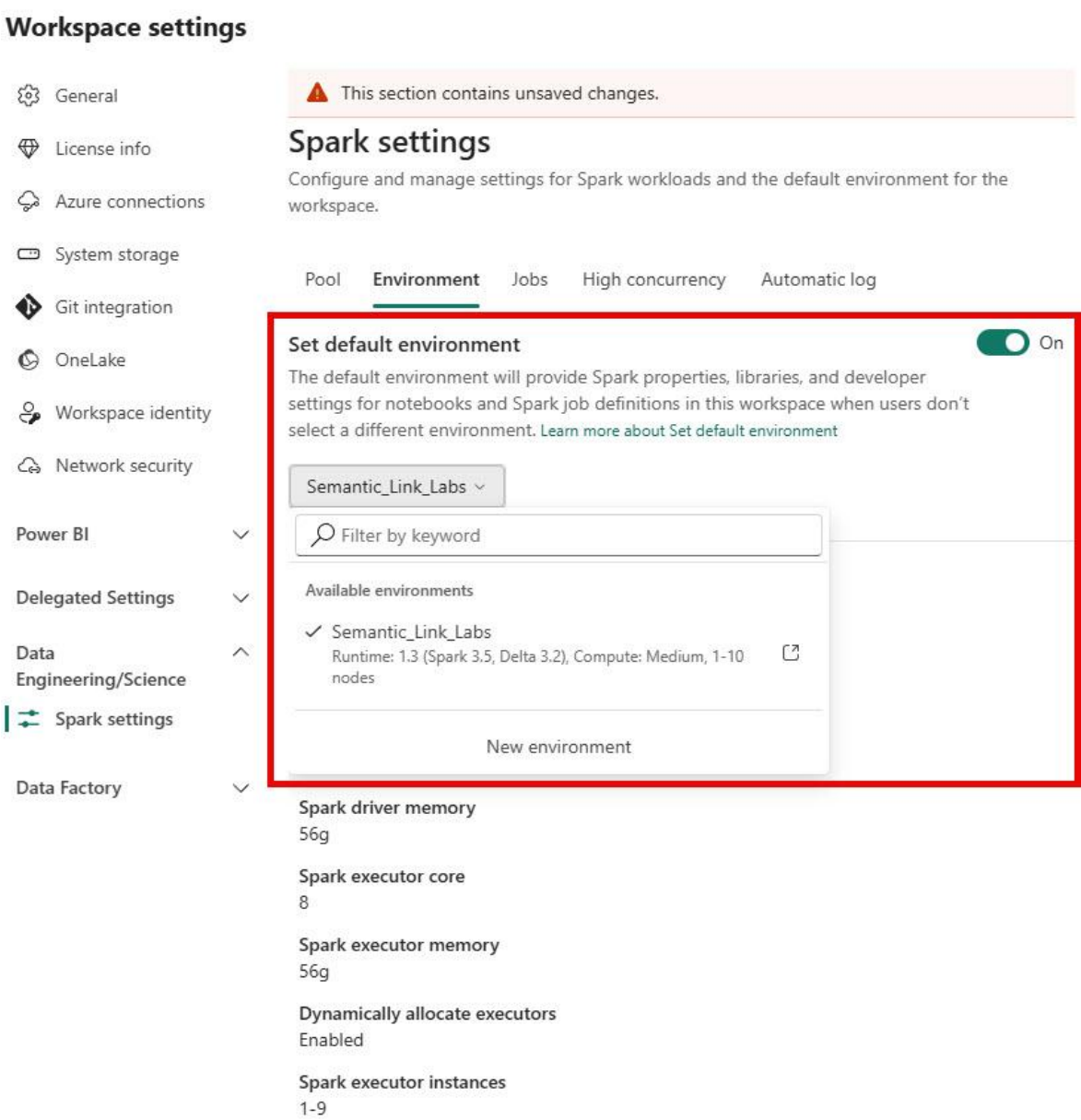
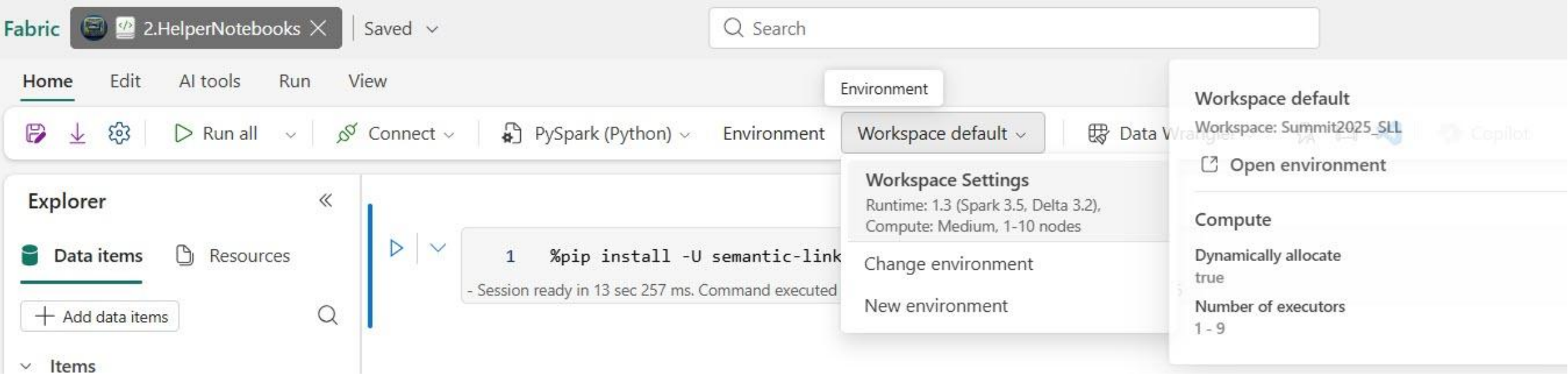
# Workspace Settings – Environment



# Set Environment - Notebook Setting



# Executing Notebook using a Pipeline

Either

- Use Environment with Semantic Link Labs
  - Microsoft recommended way
  - Only for Spark
- Python inline installation
  - Enable %pip install for pipeline, add "_inlineInstallationEnabled" as bool parameter equals True in the notebook activity parameters.

Reference:

https://learn.microsoft.com/en-us/fabric/data-engineering/library-management

# Semantic Link Labs Demo

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# Conclusion

- Your go-to solution (sooner or later)
- Ever-expanding world of capabilities
- Evolving to adapt to Fabric changes
  - For Example: Lakehouse Schemas

# Resources

- Semantic Link
  https://learn.microsoft.com/en-us/fabric/data-science/semantic-link-overview
- Semantic Link labs
  https://github.com/microsoft/semantic-link-labs
- Semantic Link Labs – Read the Docs
  https://semantic-link-labs.readthedocs.io/en/stable/sempy_labs.html

# More Resources

- Tabular Editor Training (Notebooks, Semantic Link, and Semantic Link Labs)
  https://tabulareditor.com/learn
- Sandeep Pawar
  https://fabric.guru/fabric-semantic-link-and-use-cases

# Thank you

- Jason Romans
- thedaxshepherd@gmail.com
- http://www.thedaxshepherd.com/

The DAX Shepherd

**Stay online for my live Q&A sessions**

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